

Abu Dhabi Company for Onshore Petroleum Operations Ltd(ADCO)

SHAH

SY-110

SY-110

SY01101

OH

Design: OH

Standard Survey Report

28 January, 2018

Survey Report

Company:	Abu Dhabi Company for Onshore Petroleum Operations Ltd(ADCO)	Local Co-ordinate Reference:	Well SY-110
Project:	SHAH	TVD Reference:	RTKB @ 368.6ft
Site:	SY-110	MD Reference:	RTKB @ 368.6ft
Well:	SY-110	North Reference:	Grid
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	EDMPRODR5K

Project	SHAH, SHAH		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	Nahrwan - U. Arab Emer.		
Map Zone:	Zone 39N (48 E to 54 E)		

Site	SY-110		
Site Position:		Northing:	2,535,131.00 m
From:	Map	Easting:	786,574.00 m
Position Uncertainty:	0.0 ft	Slot Radius:	13.200 in
		Latitude:	22° 54' 5.639 N
		Longitude:	53° 47' 36.515 E
		Grid Convergence:	1.09 °

Well	SY-110		
Well Position	+N/-S	0.0 ft	Northing: 2,535,131.00 m
	+E/-W	0.0 ft	Easting: 786,574.00 m
Position Uncertainty	0.0 ft	Wellhead Elevation:	ft
		Latitude:	22° 54' 5.639 N
		Longitude:	53° 47' 36.515 E
		Ground Level:	338.2 ft

Wellbore	OH				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	WMM_2005	12/31/2009	1.56	34.98	42,621.80476055

Design	OH			
Audit Notes:				
Version:	Phase:	ACTUAL	Tie On Depth:	0.0
Vertical Section:	Depth From (TVD) (ft)	+N/-S (ft)	+E/-W (ft)	Direction (°)
	-9.4	0.0	0.0	209.58

Survey Program	Date	1/28/2018		
From (ft)	To (ft)	Survey (Wellbore)	Tool Name	Description
90.6	5,062.6	GYRO 7" liner and 9-5/8" Csg. (OH)	CT_GYRO_MS	Continuous Gyro Multishot
5,062.6	5,159.0	8 1/2" HOLE SURVEY (OH)	MWD ISCWSA	MWD - Standard
5,228.7	8,700.0	6' HOLE SURVEY (OH)	MWD ISCWSA	MWD - Standard

Survey										
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)	
100.0	0.18	152.29	4,252.6	-4,458.2	-2,537.1	5,129.5	0.00	0.00	0.00	
200.0	0.25	146.73	4,352.6	-4,458.5	-2,536.9	5,129.7	0.08	0.07	-5.56	
300.0	0.34	136.97	4,452.6	-4,458.9	-2,536.6	5,129.9	0.10	0.09	-9.76	
400.0	0.37	120.82	4,552.6	-4,459.3	-2,536.1	5,130.0	0.11	0.03	-16.15	
500.0	0.27	112.46	4,652.6	-4,459.5	-2,535.6	5,130.0	0.11	-0.10	-8.36	
600.0	0.21	113.48	4,752.6	-4,459.7	-2,535.2	5,129.9	0.06	-0.06	1.02	
700.0	0.08	309.27	4,852.6	-4,459.8	-2,535.0	5,129.9	0.29	-0.14	-164.21	
800.0	0.40	302.59	4,952.6	-4,459.6	-2,535.3	5,129.9	0.32	0.32	-6.68	

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Well:	SY-110	North Reference:	Grid
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	EDMPRODR5K

Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
900.0	0.51	307.42	5,052.6	-4,459.1	-2,536.0	5,129.8	0.12	0.11	4.83
1,000.0	0.53	303.23	5,152.6	-4,458.5	-2,536.8	5,129.7	0.04	0.02	-4.20
1,100.0	0.45	299.21	5,252.6	-4,458.1	-2,537.5	5,129.7	0.09	-0.08	-4.02
1,173.0	0.54	304.12	5,325.6	-4,457.8	-2,538.0	5,129.6	0.14	0.13	6.73
SURFACE CASING									
1,200.0	0.55	301.12	5,352.6	-4,457.6	-2,538.3	5,129.6	0.11	0.03	-11.13
1,300.0	0.14	298.55	5,452.6	-4,457.4	-2,538.8	5,129.7	0.41	-0.41	-2.57
1,400.0	0.04	197.59	5,552.6	-4,457.3	-2,538.9	5,129.7	0.15	-0.10	-100.96
1,500.0	0.06	61.22	5,652.6	-4,457.4	-2,538.8	5,129.7	0.10	0.02	-136.37
1,600.0	0.06	305.75	5,752.6	-4,457.3	-2,538.8	5,129.6	0.10	-0.01	-115.47
1,700.0	0.03	248.10	5,852.6	-4,457.2	-2,538.8	5,129.6	0.05	-0.03	-57.65
1,800.0	0.10	179.35	5,952.6	-4,457.3	-2,538.8	5,129.7	0.09	0.07	-68.75
1,900.0	0.11	106.80	6,052.6	-4,457.4	-2,538.7	5,129.7	0.12	0.01	-72.55
2,000.0	0.12	84.97	6,152.6	-4,457.5	-2,538.5	5,129.6	0.04	0.01	-21.82
2,100.0	0.04	254.05	6,252.6	-4,457.4	-2,538.4	5,129.5	0.16	-0.08	169.08
2,200.0	0.12	164.53	6,352.6	-4,457.5	-2,538.5	5,129.6	0.13	0.08	-89.52
2,300.0	1.05	178.99	6,452.6	-4,458.2	-2,538.5	5,130.3	0.93	0.93	14.46
2,400.0	1.55	188.39	6,552.6	-4,460.4	-2,538.7	5,132.2	0.54	0.50	9.40
2,500.0	3.46	170.94	6,652.5	-4,464.4	-2,538.6	5,135.7	2.03	1.91	-17.45
2,600.0	7.67	170.21	6,752.0	-4,473.7	-2,537.0	5,143.0	4.21	4.21	-0.73
2,700.0	11.74	170.20	6,850.6	-4,490.4	-2,534.0	5,156.0	4.07	4.07	-0.02
2,800.0	15.28	172.22	6,947.7	-4,513.7	-2,530.6	5,174.6	3.57	3.54	2.02
2,900.0	18.34	172.38	7,043.4	-4,542.6	-2,526.7	5,197.9	3.06	3.06	0.16
3,000.0	21.81	173.64	7,137.3	-4,576.5	-2,522.5	5,225.2	3.50	3.47	1.26
3,100.0	26.03	176.47	7,228.6	-4,617.1	-2,519.0	5,258.9	4.37	4.22	2.83
3,200.0	29.06	177.26	7,317.3	-4,663.3	-2,516.7	5,297.8	3.06	3.04	0.79
3,300.0	32.73	176.69	7,403.2	-4,714.4	-2,514.0	5,341.0	3.68	3.67	-0.57
3,400.0	36.69	175.56	7,485.3	-4,771.3	-2,510.0	5,388.5	4.01	3.96	-1.13
3,500.0	40.14	175.09	7,563.5	-4,833.3	-2,505.0	5,440.0	3.45	3.44	-0.47
3,600.0	43.88	174.91	7,637.8	-4,900.0	-2,499.0	5,495.0	3.75	3.75	-0.18
3,700.0	44.30	175.00	7,709.5	-4,969.4	-2,492.8	5,552.3	0.42	0.41	0.10
3,800.0	46.41	175.64	7,780.4	-5,039.7	-2,487.1	5,610.6	2.16	2.12	0.63
3,900.0	50.28	176.11	7,846.8	-5,114.2	-2,481.8	5,672.8	3.88	3.86	0.47
4,000.0	54.39	176.60	7,907.9	-5,193.2	-2,476.7	5,739.0	4.14	4.12	0.49
4,100.0	57.69	177.47	7,963.6	-5,276.2	-2,472.4	5,809.0	3.37	3.30	0.87
4,152.0	58.68	178.00	7,990.9	-5,320.4	-2,470.6	5,846.5	2.08	1.90	1.02
PRODUCTION CASING									
4,200.0	58.64	181.82	8,015.8	-5,361.4	-2,470.2	5,882.0	6.79	-0.08	7.95
4,300.0	63.52	191.44	8,065.8	-5,447.2	-2,480.6	5,961.8	9.73	4.88	9.63
4,400.0	68.87	194.75	8,105.0	-5,536.5	-2,502.9	6,050.4	6.15	5.36	3.31
4,500.0	69.68	194.73	8,139.8	-5,627.1	-2,526.8	6,141.0	0.81	0.81	-0.02
4,600.0	72.64	194.14	8,173.7	-5,718.2	-2,550.2	6,231.8	3.01	2.96	-0.59
4,700.0	75.82	197.59	8,200.5	-5,811.0	-2,576.0	6,325.2	4.59	3.17	3.45

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Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	EDMPRODR5K

Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
4,800.0	77.39	202.15	8,222.8	-5,902.6	-2,609.3	6,421.3	4.70	1.57	4.56
4,900.0	81.08	205.91	8,242.1	-5,992.3	-2,649.0	6,518.9	5.22	3.69	3.76
5,000.0	86.07	209.58	8,254.8	-6,080.2	-2,694.8	6,618.0	6.18	4.99	3.67
5,100.0	90.23	214.53	8,255.2	-6,164.7	-2,748.2	6,717.8	6.46	4.16	4.95
5,197.0	89.59	217.69	8,255.7	-6,243.1	-2,805.2	6,814.2	3.32	-0.66	3.25
PRODUCTION LINER									
5,200.0	89.59	217.80	8,255.7	-6,245.5	-2,807.1	6,817.1	3.86	0.13	3.86
5,300.0	90.38	221.22	8,255.9	-6,322.6	-2,870.8	6,915.6	3.51	0.79	3.42
5,400.0	90.29	223.37	8,255.1	-6,396.4	-2,938.2	7,013.1	2.15	-0.09	2.15
5,500.0	89.15	224.27	8,255.5	-6,468.5	-3,007.5	7,110.0	1.46	-1.14	0.90
5,600.0	89.01	222.33	8,257.2	-6,541.1	-3,076.2	7,207.1	1.95	-0.14	-1.94
5,700.0	89.40	218.73	8,258.5	-6,617.1	-3,141.2	7,305.2	3.62	0.39	-3.60
5,800.0	88.95	216.64	8,260.0	-6,696.4	-3,202.0	7,404.3	2.14	-0.45	-2.09
5,900.0	88.96	217.22	8,261.8	-6,776.2	-3,262.3	7,503.4	0.57	0.01	0.57
6,000.0	89.00	216.75	8,263.6	-6,856.2	-3,322.3	7,602.6	0.47	0.04	-0.47
6,100.0	88.94	216.79	8,265.4	-6,936.1	-3,382.3	7,701.7	0.07	-0.06	0.04
6,200.0	89.12	217.22	8,267.2	-7,016.5	-3,441.8	7,801.0	0.47	0.17	0.44
6,300.0	89.27	220.13	8,268.3	-7,094.4	-3,504.5	7,899.7	2.92	0.16	2.91
6,400.0	88.90	220.62	8,270.1	-7,170.0	-3,569.9	7,997.7	0.61	-0.37	0.48
6,500.0	89.06	219.67	8,271.9	-7,246.6	-3,634.1	8,096.1	0.96	0.16	-0.94
6,600.0	89.23	219.62	8,273.3	-7,323.7	-3,697.8	8,194.6	0.17	0.17	-0.05
6,700.0	89.52	222.28	8,274.5	-7,399.7	-3,762.8	8,292.7	2.68	0.29	2.66
6,800.0	89.56	220.97	8,275.2	-7,473.6	-3,830.2	8,390.2	1.31	0.04	-1.31
6,900.0	89.54	221.10	8,276.0	-7,549.4	-3,895.4	8,488.4	0.13	-0.02	0.13
7,000.0	89.54	221.54	8,276.8	-7,624.5	-3,961.4	8,586.2	0.45	0.00	0.45
7,100.0	89.54	220.43	8,277.6	-7,699.9	-4,027.1	8,684.2	1.11	0.00	-1.11
7,200.0	89.57	220.05	8,278.4	-7,776.3	-4,091.6	8,782.5	0.38	0.03	-0.38
7,300.0	89.52	221.75	8,279.2	-7,851.8	-4,157.2	8,880.6	1.69	-0.05	1.69
7,400.0	89.57	221.62	8,280.0	-7,926.4	-4,223.7	8,978.3	0.14	0.06	-0.12
7,500.0	89.56	221.34	8,280.7	-8,001.4	-4,289.9	9,076.2	0.28	-0.01	-0.28
7,600.0	89.48	221.63	8,281.6	-8,076.2	-4,356.2	9,174.0	0.30	-0.08	0.28
7,700.0	89.56	221.07	8,282.4	-8,151.2	-4,422.4	9,271.9	0.56	0.09	-0.56
7,800.0	89.55	220.26	8,283.2	-8,227.1	-4,487.5	9,370.0	0.81	-0.02	-0.81
7,900.0	89.53	220.74	8,284.0	-8,303.5	-4,552.0	9,468.3	0.48	-0.02	0.48
8,000.0	89.53	221.82	8,284.8	-8,378.4	-4,618.2	9,566.2	1.08	0.00	1.08
8,100.0	89.50	220.30	8,285.7	-8,453.4	-4,684.4	9,664.0	1.52	-0.03	-1.52
8,200.0	89.50	219.82	8,286.6	-8,530.3	-4,748.3	9,762.5	0.48	0.00	-0.48
8,300.0	89.27	217.68	8,287.6	-8,607.9	-4,811.4	9,861.1	2.15	-0.23	-2.14
8,400.0	89.46	220.49	8,288.8	-8,686.2	-4,873.5	9,959.9	2.81	0.19	2.81
8,500.0	89.56	218.82	8,289.6	-8,762.7	-4,937.9	10,058.2	1.67	0.10	-1.66
8,600.0	89.45	218.83	8,290.5	-8,840.7	-5,000.5	10,156.9	0.12	-0.12	0.01
8,700.0	89.54	218.37	8,291.3	-8,918.8	-5,062.9	10,255.6	0.47	0.09	-0.46
TD									

Survey Report

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Site:	SY-110	MD Reference:	RTKB @ 368.6ft
Well:	SY-110	North Reference:	Grid
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	EDMPRODR5K

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Casing Points					
Measured Depth (ft)	Vertical Depth (ft)	Name	Casing Diameter (in)	Hole Diameter (in)	
1,173.0	5,325.6	SURFACE CASING	13.375	13.375	
4,152.0	7,990.9	PRODUCTION CASING	9.625	12.250	
5,197.0	8,255.7	PRODUCTION LINER	7.000	8.500	

Formations					
Measured Depth (ft)	Vertical Depth (ft)	Name	Lithology	Dip (°)	Dip Direction (°)
8,700.0	8,291.3	TD		0.00	

Checked By: _____	Approved By: _____	Date: _____
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